Zubair Mughal
LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

SD

COURSE 12 OF 12
CURRICULUM GUIDE

System Design for Teens

How large software systems get architected

GRADE BAND	DIFFICULTY	DURATION
Grade 10–12	Advanced	10 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

A rare, career-differentiating skill at this level: how real systems are planned, from users and APIs to databases, caching and scalability — taught conceptually and applied to a capstone design.

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Prerequisites: SQL for Teens & Beginners and Node.js for Teens & Beginners recommended.

LEARNING OUTCOMES

What Students Will Learn

- Understand the components of a typical web application architecture
- Design simple APIs and data models for a given problem
- Reason about databases, caching and scalability at a conceptual level
- Understand real engineering tradeoffs: performance, cost, complexity, reliability
- Communicate a system design clearly using diagrams
- Apply system-design thinking to a self-chosen capstone application

WHAT WE'LL USE

Tools & Technologies

Diagramming tools (e.g. Excalidraw) **Markdown documentation**

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · What Is System Design?	Real-world examples and why it matters.
02 · Clients, Servers & APIs	Architecture fundamentals.
03 · Databases at Scale	Indexing, and relational vs. NoSQL, conceptually.

MODULE	FOCUS
04 · Caching & Performance	Why and how systems get faster.
05 · Scalability Concepts	Load, and horizontal vs. vertical scaling.
06 · Reliability & Tradeoffs	Redundancy and handling failure, conceptually.
07 · Designing a System	A guided design exercise, e.g. a simple social app.
08 · Capstone Design Document	A full design document for a self-chosen application.

LEARNING BY BUILDING

Projects Students Will Build

- System Design Case Studies — Analyzing how familiar apps might work under the hood.
- Guided System Design Exercise — A structured, mentor-led design walkthrough.
- Capstone Design Document — An original system design with diagrams.

BEYOND THE SYNTAX

Skills Developed

Architectural thinking · Technical communication · Tradeoff analysis · Diagramming

PORTFOLIO OUTCOME

A polished system design document — the kind used in real technical interviews and portfolios.

RECOMMENDED NEXT STEP

The program capstone project